Statistics for Data Science and Business Analysis
Glossary

Section	Lesson	Word	Definition
1	Population vs sample	population	The collections of all items of interest to our study; denoted N.
1	Population vs sample	sample	A subset of the population; denoted n.
1	Population vs sample	parameter	A value that refers to a population. It is the opposite of statistic.
1	Population vs sample	statistic	A value that refers to a sample. It is the opposite of a parameter.
1	Population vs sample	random sample	A sample where each member is chosen from the population strictly by chance
2	Types of data	representative sample	A sample taken from the population to reflect the population as a whole
2	Types of data	variable	A characteristic of a unit which may assume more than one value. Eg. height, occupation, age etc.
2	Types of data	type of data	A way to classify data. There are two types of data - categorical and numerical.

2	Types of data	categorical data	A subgroup of types of data. Describes categories or groups.
2	Types of data	numerical data	A subgroup of types of data. Represents numbers. Can be further classified into discrete and continuous.
2	Types of data	discrete data	Data that can be counted in a finite matter. Opposite of continuous.
2	Types of data	continuous data	Data that is 'infinite' and impossible to count. Opposite of discrete.
2	Levels of measurement	levels of measurement	A way to classify data. There are two levels of measurement - qualitative and quantitative, which are further classed into nominal & ordinal, and ratio & interval, respectively.
2	Levels of measurement	qualitative data	A subgroup of levels of measurement. There are two types of qualitative data - nominal and ordinal.
2	Levels of measurement	quantitative data	A subgroup of levels of measurement. There are two types of quantitative data - ratio and interval.
2	Levels of measurement	nominal	Refers to variables that describe different categories and cannot be put in any order.
2	Levels of measurement	ordinal	Refers to variables that describe different categories, but can be ordered.
2	Levels of measurement	ratio	A number that has a unique and unambiguous zero point, no matter if a whole number or a fraction

2	Levels of measurement	interval	An interval variable represents a number or an interval. There isn't a unique and unambiguous zero point. For example, degrees in Celsius and Fahrenheit are interval variables, while Kelvin is a ratio variable.
2	Categorical variables. Visualization techniques	frequency distribution table	A table that represents the frequency of each variable.
2	Categorical variables. Visualization techniques	frequency	Measures the occurrence of a variable.
2	Categorical variables. Visualization techniques	absolute frequency	Measures the NUMBER of occurrences of a variable.
2	Categorical variables. Visualization techniques	relative frequency	Measures the RELATIVE NUMBER of occurrences of a variable. Usually, expressed in percentages.
2	Categorical variables. Visualization techniques	cumulative frequency	The sum of relative frequencies so far. The cumulative frequency of all members is 100% or 1.
2	Categorical variables. Visualization techniques	Pareto diagram	A type of bar chart where frequencies are shown in descending order. There is an additional line on the chart, showing the cumulative frequency.
2	The Histogram	histogram	A type of bar chart that represents numerical data. It is divided into intervals (or bins) that are not overlapping and span from the first observation to the last. The intervals (bins) are adjacent - where one stops, the other starts.
2	The Histogram	bins (histogram)	The intervals that are represented in a histogram.
2	Cross table and scatter plot	cross table	A table which represents categorical data. On one axis we have the categories, and on the other - their frequencies. It can be built with absolute or relative frequencies.

2	Cross table and scatter plot	contigency table	See cross table.
2	Cross table and scatter plot	scatter plot	A plot that represents numerical data. Graphically, each observation looks like a point on the scatter plot.
2	Mean, median and mode	measures of central tendency	Measures that describe the data through 'averages'. The most common are the mean, median and mode. There is also geometric mean, harmonic mean, weighted-average mean, etc.
2	Mean, median and mode	mean	The simple average of the dataset. Denoted μ.
2	Mean, median and mode	median	The middle number in an ordered dataset.
2	Mean, median and mode	mode	The value that occurs most often. A dataset can have 0, 1 or multiple modes.
2	Skewness	measures of asymmetry	Measures that describe the data through the level of symmetry that is observed. The most common are skewness and kurtosis.
2	Skewness	skewness	A measure that describes the dataset's symmetry around its mean.
2	Variance	sample formula	A formula that is calculated on a sample. The value obtained is a statistic.
2	Variance	population formula	A formula that is calculated on a population. The value obtained is a parameter.

2	Variance	measures of variability	Measures that describe the data through the level of dispersion (variability). The most common ones are variance and standard deviation.
2	Variance	variance	Measures the dispersion of the dataset around its mean. It is measured in units squared. Denoted σ^2 for a population and s^2 for a sample.
2	Standard deviation and coefficient of variation	standard deviation	Measures the dispersion of the dataset around its mean. It is measured in original units. It is equal to the square root of the variance. Denoted σ for a population and s for a sample.
2	Standard deviation and coefficient of variation	coefficient of variation	Measures the dispersion of the dataset around its mean. It is also called 'relative standard deviation'. It is useful for comparing different datasets in terms of variability.
2	Covariance	univariate measure	A measure which refers to a single variable.
2	Covariance	multivariate measure	A measure which refers to multiple variables.
2	Covariance	covariance	A measure of relationship between two variables. Usually, because of its scale of measurement, covariance is not directly interpretable. Denoted σ_{xy} for a population and s_{xy} for a sample.
2	Correlation	linear correlation coefficient	A measure of relationship between two variables. Very useful for direct interpretation as it takes on values from [-1,1]. Denoted ρ_{xy} for a population and r_{xy} for a sample.
2	Correlation	correlation	A measure of the relationship between two variables. There are several ways to compute it, the most common being the linear correlation coefficient.